



Mechatronic Sensors Trainer

Fundamental Labs – Camera

V1.0 – November 1st, 2025

© 2025 Quanser Consulting Inc., All rights reserved.
For more information on the solutions Quanser offers,
please visit the web site at: <http://www.quanser.com>



Quanser Consulting Inc. info@quanser.com
119 Spy Court Phone : 19059403575
Markham, Ontario Fax : 19059403576
L3R 5H6, Canada printed in Markham, Ontario.

This document and the software described in it are provided subject to a license agreement. Neither the software nor this document may be used or copied except as specified under the terms of that license agreement. Quanser Consulting Inc. ("Quanser") grants the following rights: a) The right to reproduce the work, to incorporate the work into one or more collections, and to reproduce the work as incorporated in the collections, b) to create and reproduce adaptations provided reasonable steps are taken to clearly identify the changes that were made to the original work, c) to distribute and publicly perform the work including as incorporated in collections, and d) to distribute and publicly perform adaptations. The above rights may be exercised in all media and formats whether now known or hereafter devised. These rights are granted subject to and limited by the following restrictions: a) You may not exercise any of the rights granted to You in above in any manner that is primarily intended for or directed toward commercial advantage or private monetary compensation, and b) You must keep intact all copyright notices for the Work and provide the name Quanser for attribution. These restrictions may not be waived without express prior written permission of Quanser.



Caution

This equipment is designed to be used for educational and research purposes and is not intended for use by the public. The user is responsible for ensuring that the equipment will be used by technically qualified personnel only. Users are responsible for certifying any modifications or additions they make to the default configuration.

Mechatronic Sensors Trainer – Application Guide

Camera Lab

Why Explore Cameras?

Cameras are key components in modern mechatronics applications. They allow us to passively obtain visual information about the environment and apply computer vision algorithms for a specific task. Applications include object inspection, facial and gesture recognition, 3D mapping, object/scene classification, etc.

Background

This lab is part of the Fundamentals labs for the Mechatronic Sensors Trainer. These labs will guide you through the sensors that are part of the board and the ones that are provided as external sensors so you can understand how to read from the sensors, what they read, when they are used, and you feel comfortable using them. This will also include highlighting datasheets to learn how to interpret them and know what to look for.

Prior to starting this lab, please review the following concept review (located in Documents/Quanser/4_concept_reviews/),

- 4_concept_reviews/Perception/**001_camera_model**
- 4_concept_reviews/Perception/**002_calibration_images**

Getting Started

In this lab, you will use the **Sensors Trainer** to learn about the basics of **Camera** sensors. The lab will guide you through camera sensor considerations as well as differences between color and grayscale images.

Before you begin this lab, ensure that the following criteria are met:

- Make sure the Sensors Trainer has been setup and tested. See the **Quick Start Guide (Quanser/2_quick_start_guides/sensors_trainer)** for details on this step.
- You are familiar with the basics of the programming language you are using for this lab.
- Have your development environment ready,
 - o If using Python, we recommend using Visual Studio Code.